TECHNISCHE UNIVERSITÄT DRESDEN

Faculty of Computer Science Institute of System Architectures, Chair of Privacy and Security

Peer-to-Peer Systems | Prof. Tschorsch

Peer-to-Peer Project | Summer 2026

Objective

The goal of this project is to design and implement a **peer-to-peer (P2P) application** over the course of the semester. You are expected to apply concepts from the lecture to build a decentralized system with a clear focus on protocol design, robustness, and scalability.

You determine the topic of your project yourself. For example, you may develop a new P2P application (e.g., file sharing, messaging, or distributed storage) or extend an existing program with a peer-to-peer component.

Students work in groups of three to five. The workload (technical complexity) of the project are expected to scale with the number of group members. Each group must register in OPAL.

Requirements

Your project should:

- function without a central authority
- implement the necessary peer-to-peer protocols and algorithms at the application layer (typically based on TCP or UDP sockets)
- in principle support any number of participants
- tolerate peer failures such that the system remains operational
- be functional over the Internet (no multicast, no broadcast)
- ignore bootstrapping and NAT traversal

Projects will be evaluated based on their originality, the technical sophistication of the solution, the quality and robustness of the implementation, and the clarity of the final presentation.

Project Timeline

By **May 18**, each group submits two slides (PDF, 16:9): one slide describing the project objective(s) and one slide outlining the technical challenge(s). On **May 19**, each group presents its project idea in a 5-minute presentation, followed by a discussion of scope and feasibility. The system is then implemented over the remainder of the semester. On **July 14**, each group presents its final results including a demo of the project.

- **May 18**: Submission of project proposal slides
- **May 19**: Project idea presentations
- **July 14**: Final project presentations